Week 28: Flat folds

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This view repeats, by problem, the answers of the print guide, week-28-facilitator.pdf, whose source is editable/facilitator-src/build_guide.py; its page references are to that guide. The overview, preparation, launch, proofs and hints are there. Change both together; check_guide_copy.py compares them.

Problems and routes

Problem	K–1	2–3	4–5	route
Close the four-ray disk in four ways	P1			first
Which printed folds can you make?	P2			first
Every fold with ray 1 a mountain	P3			return
Finish three disks from one row of tabs	P4			return
Half-turns from seven wedges		P1		first
Which four-ray disks close flat?		P2	P1	first
Do the shaded and white pairs make half-turns?		P3		first
Add one ray to make a flat pattern		P4		return
Patterns with an odd number of rays			P2	first
Every other sector totals a half-turn			P3	first
Every flat labeling of the four-right-angle disk	P5	P5	P4	return
Routes through the eight folds	P6			return
Rule out the remaining labelings			P5	return
Orders of six wedges with alternating half-turns		P6	P6	return

First route. K–1 P1–P2; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P3–P6; Grades 2–3 P4–P6; Grades 4–5 P4–P6.

Key, one entry per problem

Close the four-ray disk in four ways

K–1 P1 · first route

Idea. The four-right-angle disk folds flat in several different mountain-valley labelings, each with three of one letter and one of the other.

Answer: any four different words in the catalog. For example A,B,C,D are all valid. Four drawings of the same labels in different physical orientations are not four numbered-ray solutions. Every word has a construction by page 3; page 6 proves that there are no others.

Which printed folds can you make?

K–1 P2 · first route

Idea. A four-right-angle labeling folds flat exactly when one ray differs from the other three.

The exact mathematical choices are A,C,E. The worksheet asks children to circle each they make; a mathematically valid case that they have not made should remain a physical attempt in progress. Rule out B,D,F by the complete four-angle classification, not by repeated frustration.

Every fold with ray 1 a mountain

K–1 P3 · return route

Idea. Fixing one label splits the eight right-angle folds into cases: ray 1 is the odd one out, or one of the other rays is.

All four answers: MMMV, MMVM, MVMM, MVVV. If M is the majority, its single V can occupy ray 2,3 or 4. If M is the minority, ray 1 is the only M. Those cases exhaust the catalog while keeping ray 1 fixed.

Finish three disks from one row of tabs

K–1 P4 · return route

Idea. Every flat right-angle disk has an odd number of M labels, so three disks together use an odd number of them.

Answer: all three disks can close in rows A and C; row B is impossible. Six tabs must fill all six blanks, with none borrowed between rows. Printed letters stay fixed.

Every legal right-angle disk has an odd number of M labels, either one or three. Across three legal disks the total number of M labels is odd. The printed letters contribute three M labels; row B adds three more, giving an even total of six. This contradiction rules out every placement of its tabs.

Half-turns from seven wedges

Grades 2–3 P1 · first route

Idea. Groups of wedges that fill a straight angle exactly are the half-turns the closing test needs; organising by the largest piece finds all ten.

All ten groups: AG, BG, CF, DF, ABF, ACE, ADE, BCE, BDE, ABCD. Order within a half-turn does not create a new group; A and B remain distinct letters even though their angles agree. Every listed group sums to 180°.

Completeness: with G=150° the remaining 30° is A or B. With F=120° and no G, the remaining 60° is C, D or AB. With E=90° and neither F nor G, the remaining 90° is one of A,B plus one of C,D, giving four groups. With none of E,F,G, all of A,B,C,D are needed. This gives 2+3+4+1=10 and covers every largest-piece case.

Which four-ray disks close flat?

Grades 2–3 P2 · Grades 4–5 P1 · first route · wording differs by band

Idea. A disk whose alternating sectors do not total a half-turn cannot fold flat; the three disks that pass the test each have a flat fold.

A, C and D can close; B cannot. One valid word for A is VVVM by two book folds; the same word works for C by the adjusted book-fold construction. For D use MVVV with the layer certificate on page 4. Other successful assignments can be accepted if an intact disk uses every listed ray and no extra crease.

B is impossible by 135° versus 225°, using the necessity proof on page 5.

Grades 4–5: The printed task deliberately permits a question mark when a child has not found a witness. Honor that recording: do not replace the child's evidence with an asserted success.

Do the shaded and white pairs make half-turns?

Grades 2–3 P3 · first route

Idea. On a four-ray disk, the alternating pairs of sectors either both make exact half-turns or one falls short by what the other exceeds.

Disk	Clockwise sectors 1,2,3,4	Shaded 1+3	White 2+4	Both half-turns?
A	90°,90°,90°,90°	180°	180°	Yes
B	45°,90°,90°,135°	135°	225°	No
C	60°,120°,120°,60°	180°	180°	Yes
D	30°,60°,150°,120°	180°	180°	Yes

For B, the shaded pair falls short by 45° and the white pair exceeds a half-turn by 45°. These are exact mismatches in the ideal pattern; imprecise cutting may blur the observation but cannot change the angle sums. The last sentence asks for this: B is the disk that could not close in Problem 2, and its pairs cannot both make a half-turn.

Add one ray to make a flat pattern

Grades 2–3 P4 · return route

Idea. Given rays at 0, a and 180 degrees, the only fourth ray that balances the alternating sectors is at 360 minus a degrees.

Case	Given rays clockwise from top	Add ray at	Final sectors clockwise
A	0°,90°,180°	270°	90°,90°,90°,90°
B	0°,60°,180°	300°	60°,120°,120°,60°
C	0°,30°,180°	330°	30°,150°,150°,30°
D	0°,120°,180°	240°	120°,60°,60°,120°

Each completion is a two-book-fold model of page 3 with the second fold at a; VVVM is a witness.

Patterns with an odd number of rays

Grades 4–5 P2 · first route

Idea. Each fold flips which face is up, so a flat fold around one point needs an even number of creases.

Case	Consecutive angles	Number of active rays	Answer
A	120°,120°,120°	3	Impossible
B	60°,60°,60°,90°,90°	5	Impossible
C	30°,30°,60°,60°,60°,60°,60°	7	Impossible

The orientation-parity proof on page 5 covers all three. The sectors sum to 360° in each case, so a missing wedge is not the obstruction.

Every other sector totals a half-turn

Grades 4–5 P3 · first route

Idea. Following the folded rays around the vertex gives an alternating sum that must be zero, so odd-position and even-position sectors each total 180 degrees (Kawasaki-Justin, necessity).

Case	Odd sectors	Even sectors	What the proof establishes
A	45+90=135°	90+135=225°	Ruled out
B	30+60+90=180°	30+60+90=180°	Not ruled out
C	30+30+90=150°	30+60+120=210°	Ruled out
D	4×45=180°	4×45=180°	Not ruled out

Answer: rule out A and C. Page 5 gives the complete requested proof, including why an apparent full-turn ambiguity cannot produce $\pm360°$. In B and D, passing this test does not validate a particular M/V choice; none is prescribed here.

Every flat labeling of the four-right-angle disk

K–1 P5 · Grades 2–3 P5 · Grades 4–5 P4 · return route · wording differs by band

Idea. The four-right-angle disk has exactly eight flat labelings: one exceptional ray in any of four positions, labelled M or V.

Full possible list: MMMV, MMVM, MVMM, VMMM, MVVV, VMVV, VVMV, VVVM.

K–1: Organize them by the exceptional ray and by whether its label is M or V: four positions times two types. The necessity proof is on page 6 and the construction is on page 3.

Grades 2–3: Keep ray 1 at the top; rotations count as new only when the numbered-ray labels change. A full justification has two parts: all listed words work by construction, and all others are ruled out by page 6.

Grades 4–5: The page asks for as many as children can find; each fold is its own witness. It does not yet require them to know the list is complete. For each state, use the two-book-fold construction, rotate the exceptional ray, and if needed reverse all folds.

Routes through the eight folds

K–1 P6 · return route

Idea. Two flat right-angle labelings always differ in an even number of rays; only the four complementary pairs differ in all four.

Two valid routes: each visits all eight printed states exactly once, uses the required endpoints and changes exactly two ray labels at every step.

Route ending H	Changed ray numbers	Route ending D	Changed ray numbers
A→B	3,4	A→B	3,4
B→C	2,3	B→C	2,3
C→D	1,2	C→E	3,4
D→F	3,4	E→F	1,2
F→E	1,2	F→G	2,3
E→G	1,3	G→H	3,4
G→H	3,4	H→D	2,3

Complete move rule: any two distinct catalog states are two changes apart except the complementary pairs A-H, B-G, C-F and D-E, which are four changes apart.

Rule out the remaining labelings

Grades 4–5 P5 · return route

Idea. Stacking the four right-angle sectors shows that every flat fold has exactly one exceptional ray, which rules out the other eight of the sixteen labelings.

Printed case	Word	All cases represented by rotation
A	MMMM	MMMM
B	VVVV	VVVV
C	MMVV	MMVV, MVVM, VVMM, VMMV
D	MVMV	MVMV, VMVM

None of A-D can close flat. Page 6 proves that every legal four-right-angle stack produces exactly one exceptional ray. Consequently four equal labels and every 2-and-2 arrangement are impossible. The table shows that rotations of the printed cases cover all eight invalid words.

Orders of six wedges with alternating half-turns

Grades 2–3 P6 · Grades 4–5 P6 · return route · wording differs by band

Idea. With wedges 30, 30, 60, 60, 90 and 90 degrees, each alternating triple must be one of each size, which gives 36 rooted orders and 12 starting with 30 degrees.

Grades 2–3: Here A=30°, B=60°, C=90°, and each letter appears twice. Four sample answers are **AABBCC, AABCCB, AACBBC, AACCBB**. Each word is read clockwise from the top ray; swapping the two physical copies of a letter makes no new order. There are 36 rooted letter orders in all, although only four are requested.

Grades 4–5: Answer: exactly twelve clockwise orders start with 30° and give two alternating sums of 180°. The first sector lies immediately clockwise after the marked top ray. Equal copies are indistinguishable; reflections and rotations are not identified after this starting convention is fixed.

The first 30° fills one odd slot. The other odd slots 3 and 5 hold 60°,90° in two possible orders. The even slots 2,4,6 hold 30°,60°,90° in $3\times2\times1=6$ orders. The choices are independent, giving $2\times6=12$.